ANALYTICAL CUME
APRIL 22, 2005

Water is an extremely interesting molecule in the solar system and beyond, with considerable scientific significance.

Name the methods that are used to detect water in the space and planetary environments below, describe the principles of their operation, practical details of their operation, and some recent results. The environments are;

(a) The interstellar medium
(b) The icy moons of the outer solar system
(b) The surface of Mars
(c) The subsurface of Mars
(d) The surface of an asteroid
(e) In a meteorite

Credit will be given for content, writing quality, quantitative arguments, diagrams and citations. Approximately equal credit will be given for each of the objects applications (a)-(e).